

Solar panels increase output in Delhi

The lockdown has caused clearer skies in Delhi and as a result there has been an 8 percent increase in the amount of power generated by solar panels installed in the city. While scientists previously did suspect that clearer skies would increase the output of solar panels, for the first time researchers have concrete data to prove by how much the power generation capacity improves. Normally, the difference is hard to measure because of the many factors that influence the power generation of solar panels, from dust to weather conditions. The same team has been monitoring the power output as well as pollution levels in the city for years, and back in 2013 had studied the impact of smoke plumes



that had drifted into the city from forest fires in southeast Asia. The team also used the air quality data collected from the US Embassy in

Delhi, which kept an hourly record of the concentration of particulate matter in the air. After the lockdown, pollution levels in the city went down by 50 percent. The findings apply all around the world, and indicate the potential capacity of solar panels to produce energy, if pollution levels linked to emissions are reduced across the planet.

<https://digit.in/pwrpnl>

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THIS MONTH IN SCIENCE:

This month learn about some climate change hotspots around the world, we explore DNA Origami, we look at the development of surgical adhesives, and more.

Bacteria with metal diet discovered

For over half a century, scientists had predicted that there might be microbes that derive their nutrition from eating metal but had not



actually found such a life form until now. Microbiologists from Caltech have identified the first such bacteria known which derive their sustenance from manganese. The bacteria uses

the manganese to convert carbon into biomass through a process known as chemosynthesis. The researchers were not even trying to find the bacteria,

but came across it by accident, when a glass with manganese used for other experiments was left soaked with water in a scientist's office. Later, he found that the glass jar was

coated with a dark substance, oxidised manganese, which was the action of the bacteria that had originated in the tap water. Researchers now speculate that the newly discovered bacteria or its relatives are responsible for the long standing mystery of manganese nodules that dot the seafloor. These nodules are found all over the world's oceans, but nobody knew where they came from so far.

<https://digit.in/mtlmcrcb>



Tetraquark discovered

While the detection of such particles have been theoretically predicted, scientists at CERN have detected a tetraquark, which is a hadron made up of four quarks instead of the usual two or three. The finding expands our understanding of quarks and opens up the possibility of finding more such particles.

<https://digit.in/ttraqrk>



Elephant deaths in Botswana

Over 280 elephants have died in Botswana because of some strange cause, and scientists are in the process of investigating the cause. The tusks are intact, so it is not poisoning by poachers. Instead, the cause is suspected to be a natural toxin or some killer microbes.

<https://digit.in/elephded>



Quantum rainbow

A random walk is used to cover an area and is used in computing to search through databases. In a quantum random walk, a single agent can move in multiple directions, following several trajectories at the same time, a property that can be used to make searches faster.

<https://digit.in/qrwalk>



VLT captures exoplanets

The European Southern Observatory has used the Very Large Telescope to directly image a pair of giant exoplanets in orbit around a Sun like star, which is the first image of its kind. The two planets are 6 and 14 times as massive as Jupiter in our own Solar system.

<https://digit.in/vltexo>